Microwave reheating of e-beam-pumped excimer plasmas

observable, real, and marginal — an in-silico study across three independent models

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Internal draft — program data-rights review before wider circulation. All numbers regenerate from recorded run artifacts.

The question, and how it was asked

Hybrid pumping: keep the e-beam as primary ionizer, add microwave power (during the pulse or in the afterglow) to reheat the electrons and harvest more excimer formation. Does it help?

- Twin pairs: identical runs $\pm$ a prescribed-power microwave window. Marginal $m = \Delta E_L / E_{\mathrm{MW}}$; blended $\Delta\eta$ with the microwave joules in the denominator; fluorescence E_{fl} = the experimental observable.
- Three independent electron treatments: 26-species 0-D engine on LoKI-MC swarm tables; the same engine on PIC/MCC-measured rates; an independently written 1-D discharge-pumped fluid model (two-term Boltzmann).
- Decidability ladder: a sign counts only if it survives numerics $\rightarrow$ rate-tier swap $\rightarrow \times\frac{1}{3}..\times 3$ rate brackets $\rightarrow$ the independent model. Otherwise it is reported *undecided*, with the measurement that decides it.

The answer in three lines

1. **Reheating is a robust observable.** The gain-proportional fluorescence rises in **8/8** benchmarks, on both rate tiers, and in the independent model ($+1.5\%$ to $+12\%$).
2. **One laser-positive channel is real** — afterglow reheat of a high-margin, still-extracting cavity: $\sim +10\%$ per microwave joule, surviving **41/41** rate brackets, tier swap (sibling cell), 1-D replication, and a **16.5**σ placebo-calibrated artifact test.
3. **Exploiting it is very difficult.** No benchmark moves its blended efficiency by more than **0.006 percentage points** at program-representative coupling; five quantitative walls stand between "real" and "useful."

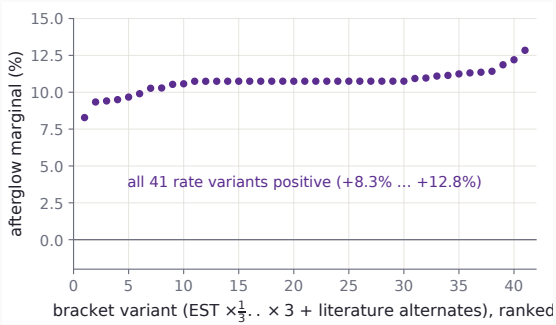
1. The observable: fluorescence responds everywhere — efficiency does not

case	η_0	afterglow m	$\Delta\eta$ (pp)	ΔE_{fl}
high-power cell, 2 atm	1.54%	−2.8%	−0.0055	+0.48 mJ
X-band chamber column	2.16%	−10.4%	−0.0036	+2.15 mJ
Ar-optimized	4.95%	−2.1%	−0.0056	+5.12 mJ
Ne-optimized	6.19%	+10.0%	+0.0024	+1.30 mJ
KrF anchor (1985)	0.70%	−0.0%	−0.0003	+0.47 mJ
ArF anchor, 1.09 atm	6.13%	−1.3%	−0.0031	+14.8 mJ

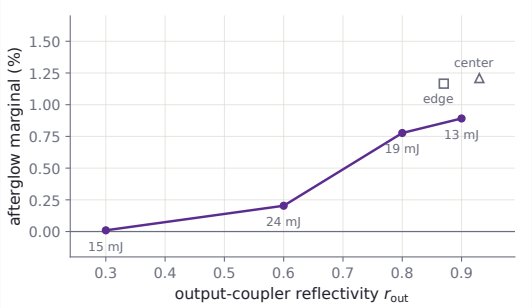
The trap for experiment design: the diagnostic (fluorescence $\propto$ gain) rises *even where laser output falls* — six of eight afterglow cases. A fluorescence increase is not an efficiency statement.

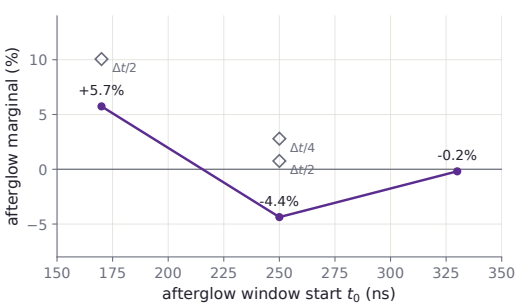
2. The real channel: late extraction, not afterglow harvest

- Every direct excimer-formation row is heavy-particle; heating the electrons cuts *both* supply lines (F^- $\times 0.13$, precursor* $\times 0.37$).
- The one positive case gains by late extraction: the high-margin cavity is still draining while the window runs ($\Delta E_L \gg \Delta E_{fl}$).
- Sign survives **all 41** rate-bracket / literature variants ($+8.3\%$ to $+12.9\%$) $\rightarrow$
- Cross-framework witnesses: kinetic tier $+9.5\%$ (sibling cell); 1-D model $+10.1\%$ at its finest timestep.



2. Verified in 1-D: the mechanism's own fingerprints





- **Margin (left):** marginal rises monotonically with r_{out} while the *absolute* output peaks at 0.6 — it tracks photon lifetime, not laser performance. Edge/center/uniform deposition: where is second-order.
- **Timing (right):** $+5.7\% \rightarrow +10.1\%$ under Δt refinement at full drain overlap; decays to ≈ 0 past the drain (the -4.4% dip was a Δt artifact — caught by refinement). When is first-order.

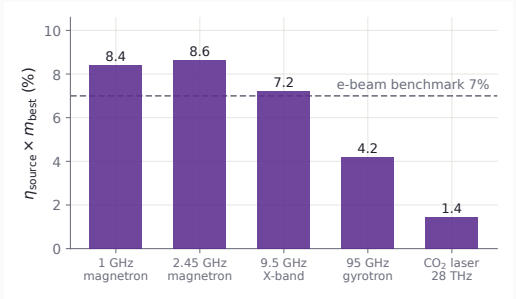
2. Not a numerical artifact: a placebo-calibrated 16.5σ

- Per solver setting (7: tolerance decades, atol, step cap, and the *independent Radau integrator*), the twin runs three ways: off / 500 kW signal / 1 W placebo — the placebo twin difference calibrates the numerical differencing noise *in situ*.
- Exact Student-t discovery: $p_0 = 1.7 \times 10^{-61}$ (13 dof) $\Rightarrow Z = 16.5\sigma$; 2×10^5 null toys: zero exceedances; 95% $\mathrm{CL_s}$ band $[8.5980, 8.5982]$ mJ excludes zero by 1.4×10^5 s.e.
- Worst single bracket variant alone: $z \approx 4 \times 10^4$; the noise would need to be wrong by $\times 4 \times 10^4$ to drop to 3σ. Conservative by construction (the placebo carries real small-signal physics). Self-testing machinery + negative control.
- 1-D: differencing floor measured by a power ladder (8×10^{-4} mJ) — the signal sits $\times 300$ above it.

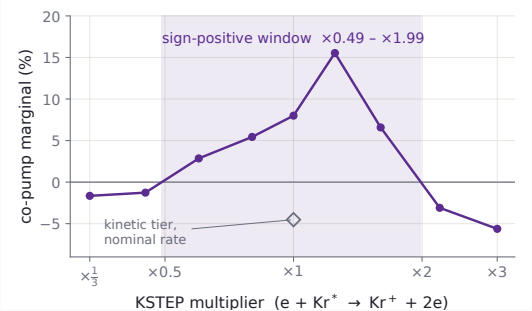
Scope: rejects a numerics-noise origin. Shared model error is covered separately: two unrelated codebases agree, energy ledger closes to 10^{-13}, 197-test suite, bracket program.

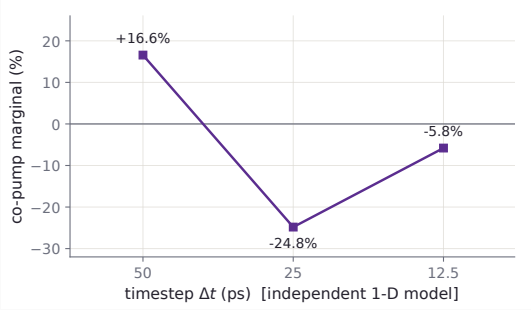
3. Why it is very difficult: five walls

1. **The ceiling:** $|\Delta\eta| \leq 0.006$ pp everywhere — the marginal must beat the cell's own η to pay.
2. **Conductivity pinning:** the afterglow settles at $\lesssim$2 Td at *any* power (to 1 GW); the useful channel needs 20–30 Td.
3. **Supply-chain inversion:** heating cuts both excimer feedstocks (KrF and ArF).
4. **Wall-plug (right):** frequency-blind below 0.3 THz; only magnetrons $\times$ the fragile $+12\%$ clear the bar ($=$ the program-stated $\sim$7% current e-beam wall-plug). The 2-mm skin makes uniform coupling an upper bound.
5. **The during-pulse sign:** one uncertain rate wide (next slide).



3. The during-pulse sign: one rate wide, and Δt-fragile





- **Left (0-D):** co-pump sign positive *iff* KSTEP ($e + \mathrm{Kr}^* \rightarrow \mathrm{Kr}^+ + 2e$) lies within $\times 0.49$–$\times 1.99$ of nominal; current knowledge ($\times 3$) straddles both crossings $\Rightarrow$ undecidable today, decided by a factor-2 rate determination.
- **Right (1-D):** the coarse-Δt +16.6% dies under refinement — small-or-negative at fine steps.

What to do with this

- **Measure one rate**: Kr* stepwise ionization to a factor of 2 (literature adjudication / driven PIC sampling / experiment) decides the co-pump sign.
- **Run the twin-shot experiment** in a high-margin cell: window on/off at fixed fill; scanning the window delay traces the drain curve — the mechanism's direct signature. Shot counts follow from measured twin variances.
- **Close the denominator**: quoting any measured marginal needs the *absorbed* microwave energy — a full-wave deposition map (the 2-mm skin vs the cell), not forward power.
- **Perspective**: the levers that actually move efficiency remain conventional — F_2 fraction, pressure, optics, buffer: a measured ladder 1.54% $\rightarrow$ 6.19%, roughly $10^3\times$ any electromagnetic effect found here.

Summary

1. Reheating works as an observable: fluorescence responds in every configuration, in every model. The diagnostic will see the microwave — do not mistake it for efficiency.
2. The afterglow gain in a still-extracting cavity is real: $\sim +10\%$ per microwave joule, bracket-proof (41/41), 1-D-verified ladder by ladder, and 16.5σ against the numerics-artifact null.
3. Converting it to device efficiency is very difficult: ≤ 0.006 pp at feasible coupling, five quantified walls, and the one larger channel hangs on a single rate known to $\times 3$.

One rate to measure, one regime to test, one observable not to be fooled by.

Backup: the full benchmark sweep (swarm tier, corrected engine)

		afterglow		co-pump	
case	η_0	m	$\Delta\eta$ (pp)	m	$\Delta\eta$ (pp)
program class, 3 atm (Ar)	no lasing	—	—	—	—
program class, 3 atm (Ne)	no lasing	—	—	—	—
high-power cell, 2 atm	1.54%	−2.8%	−0.0055	+12.5%	+0.0029
X-band chamber column	2.16%	−10.4%	−0.0036	−1.9%	−0.0004
Ar-optimized	4.95%	−2.1%	−0.0056	+1.3%	−0.0010
Ne-optimized	6.19%	+10.0%	+0.0024	+5.1%	−0.0002
KrF anchor (1985)	0.70%	−0.0%	−0.0003	+3.0%	+0.0002
ArF anchor, 1.09 atm	6.13%	−1.3%	−0.0031	+5.7%	−0.0001

Kinetic tier (5/8): re-running on the independently derived PIC-measured rate set flips the *signs* of these few-percent marginals (HYPER co-pump $+12.5 \to -2.7\%$; Ar-opt afterglow $-2.1 \to +9.5\%$) while the ≤ 0.006-pp magnitude ceiling holds on both sets (max kinetic $|\Delta\eta|$ 0.0036 pp). "Sign not tier-stable" means exactly that — the answer's sign depends on which rate set you believe, not that a bug is suspected. Full record: `mw_reheat_draft.pdf` + REBUILD CHANGELOG [0.1.20]–[0.1.33].

Backup: the arithmetic of a blended gain, derated

With and without, exactly (Ne-optimized cell): without microwave $\eta_0 = 7.773\,\mathrm{J}/125.49\,\mathrm{J} = 6.194\%$; with the afterglow window $\eta = 7.781\,\mathrm{J}/125.57\,\mathrm{J} = 6.197\%$. To first order (exact to the digits shown):

$$\Delta\eta \;\simeq\; (\eta_c\, m - \eta_0)\,\frac{E_{\mathrm{RF}}}{E_{\mathrm{tot}}}, \qquad m = +10.0\%/\text{absorbed J}, \;\; \eta_c = \text{absorbed fraction}.$$

- Intrinsic gain needs $\eta_c > \eta_0/m = 62\%$ absorbed.
- Wall-plug gain needs $\eta_{\mathrm{src}}\eta_c\, m > 7\%$: a 72% magnetron needs $\eta_c \geq 97\%$; X-band and gyrotron need $> 100\%$ — excluded at any realistic coupling (at an 8% bar, excluded outright).
- Fresnel reflection off the collisional plasma face ($\nu \approx 2 \times 10^{12}$ s^{-1}), at $n_e = 10^{19}/10^{20}/10^{21}$ m^{-3}: 2.45 GHz: 5/42/76%; 9.5 GHz: 0/17/58%; 95 GHz: 0/0/20%.

Demonstration point (intrinsic accounting): Ne-opt cell (η_0 6.2%, margin $\sim$19), afterglow window beam-off+10 ns $\times$ 150 ns, 95 GHz launch ($\eta_c \approx 0.75$–0.9 plausible; the low source efficiency is irrelevant to intrinsic bookkeeping).

Expected: $\Delta\eta \approx +0.001$ to $+0.04$ pp for $E_{\mathrm{RF}} = 0.1$–2 J — an mJ-class, 16.5σ-measurable ΔE_L, *not* a device lever.